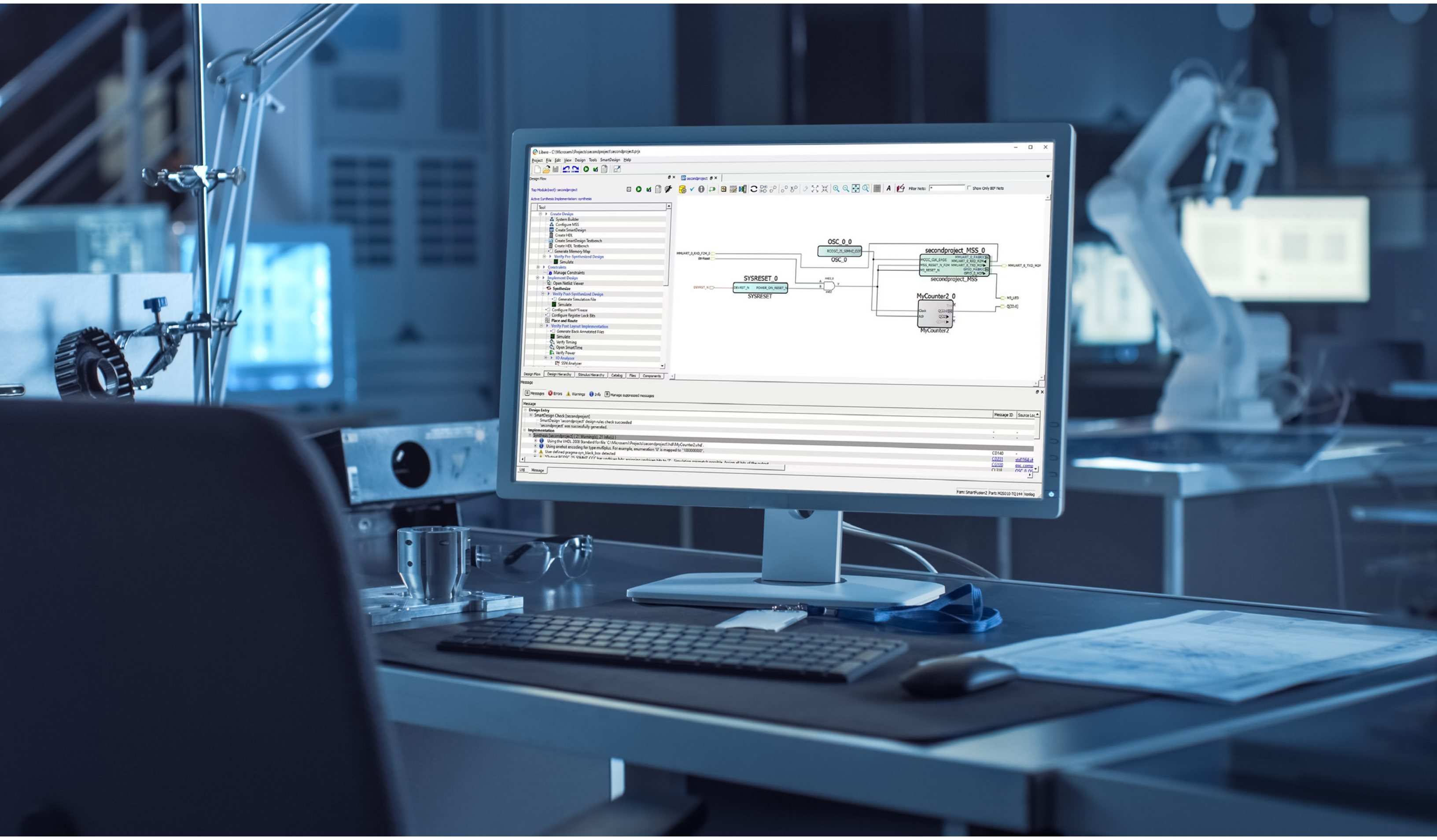
Libero® SoC Design Suite

Unleash Your Design Potential



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Overview

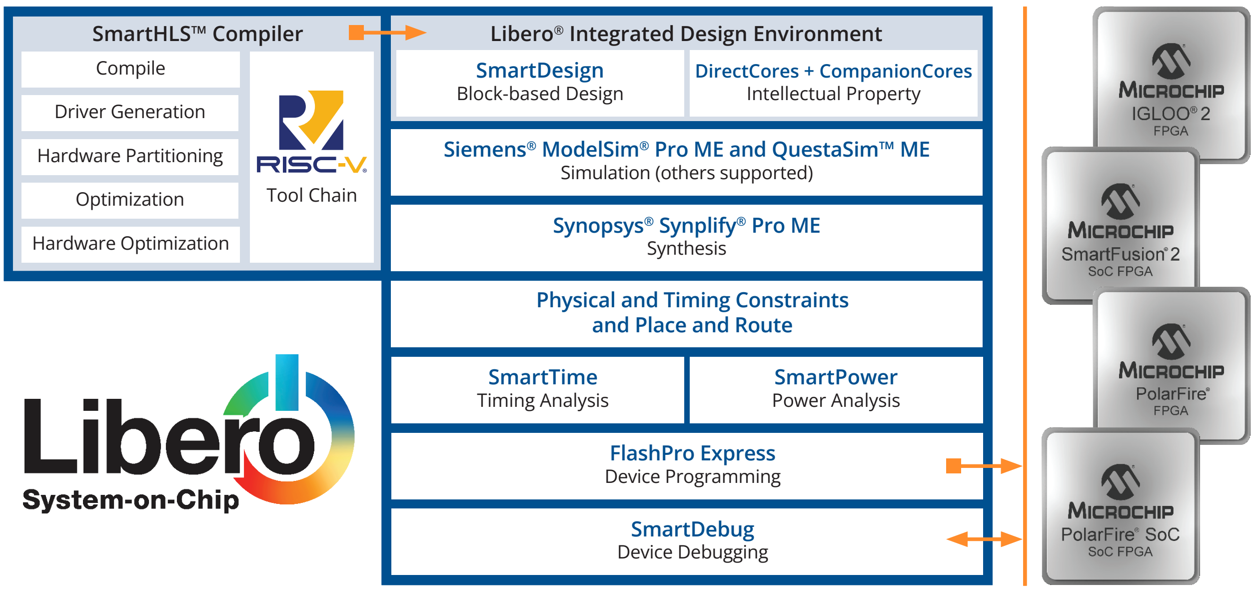
Libero® SoC Design Suite is the ultimate toolset for your FPGA design journey. This user-friendly, all-in-one software is crafted to support designers who are tackling complex FPGA and SoC projects. With its advanced Libero SoC and SmartHLS™ compiler tools, the suite simplifies your entire design process—from concept to implementation—helping you to complete projects efficiently and with confidence.

Key Features

Libero SoC Design Suite provides an integrated design environment for Register Transfer Level (RTL) design entry through device programming, a rich Intellectual Property (IP) library, and targeted reference designs and development kits. Libero SoC Design Suite allows you to:

- Seamlessly manage your entire design flow with a single, unified platform
- Achieve optimal performance and resource utilization with state-of-the-art synthesis and simulation tools
- Access a rich library of pre-verified IP cores to accelerate your design process
- Identify and resolve issues quickly with comprehensive debug and analysis capabilities
- Design for a wide range of FPGA and SoC devices, ensuring flexibility and scalability
- The SmartHLS compiler converts software algorithms into hardware designs, speeding up development and optimizing FPGA implementations for faster time to market and efficient performance. Some of its key benefits include:
- Higher productivity from design cycles that are 2.5x faster than prior versions; accelerated C/C++ algorithms written by software engineers and fewer lines of code
- Better design quality enabled by easier-to-understand and maintain C/C++ code
- Portable and flexible across Microchip's FPGA architectures, with auto partition between programmable fabrics and embedded processors
- Embedded software design is 2x to 10x faster than prior versions
- Faster debug with hardware/software co-verification

Figure 2—Libero SoC Design Suite Block Diagram



Libero SoC Design Suite Accelerates PolarFire® FPGA and SoC Designs

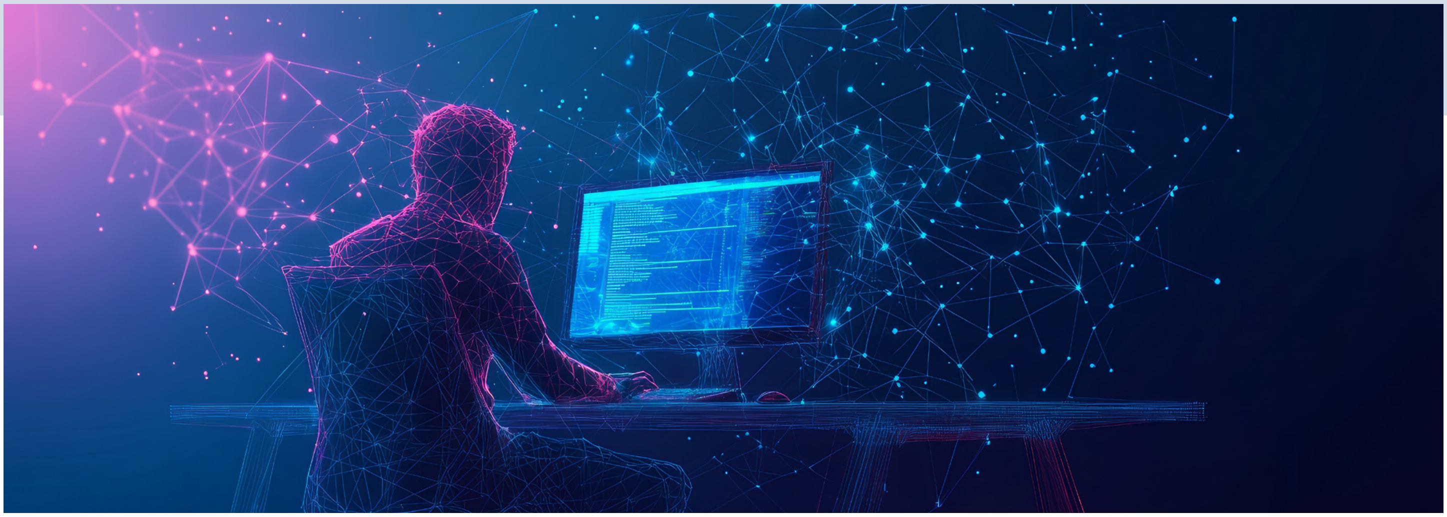
Harness the full potential of your designs with our energy-efficient, secure and highly reliable FPGAs, which are seamlessly supported by Libero SoC Design Suite. Designed for aerospace, defense, automotive and industrial applications, our FPGAs deliver exceptional energy efficiency and advanced security features to meet the most demanding standards. Libero SoC Design Suite offers an intuitive, powerful platform to optimize your FPGA projects, enabling faster development cycles while ensuring top-tier performance and reliability.

Functional Verification

Libero SoC Design Suite integrates industry-leading ModelSim® and QuestaSim™ simulators, offering advanced functional verification for FPGA designs using Verilog, SystemVerilog, mixed language and VHDL.

Performance-Optimized Floor Planner

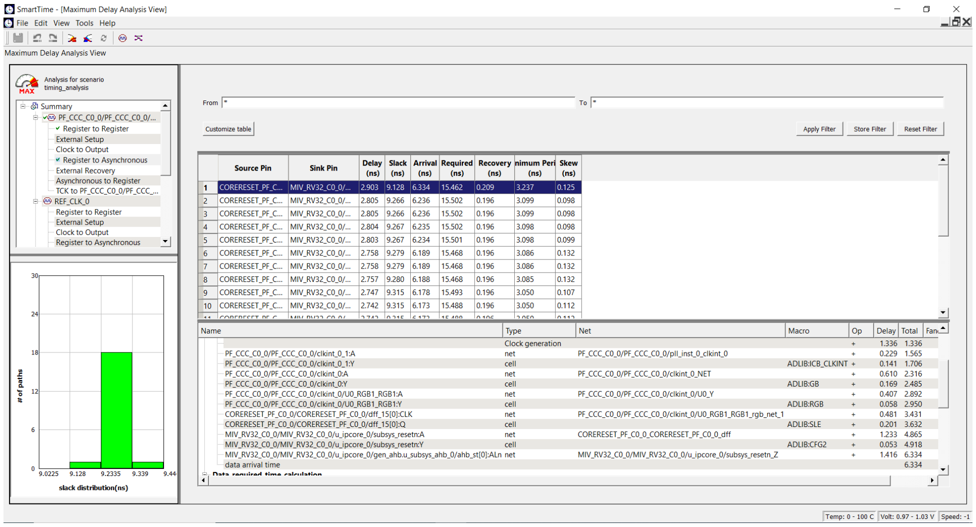
Chip Planner is an interactive tool for floorplanning, which enables advanced physical design exploration, optimization capabilities and seamless integration with constraint management tools within Libero SoC Design Suite. Floorplanning facilitates efficient design closure and accelerates time to market for FPGA-based products and solutions.



In-Depth Timing Assurance

Our advanced Static Timing Analysis (STA) ensures your digital designs meet the highest standards for timing accuracy and efficiency. Leveraging cutting-edge algorithms, our STA tools deliver precise insights into your design's timing, enabling you to identify and resolve potential issues early.

Figure 1—SmartTime Maximum Delay View



Strategic Power Analysis for Improved Design Efficiency

SmartPower makes it possible for you to achieve significant power savings, optimize energy efficiency and meet stringent power requirements while maintaining the performance and functionality of your FPGA designs. It features:

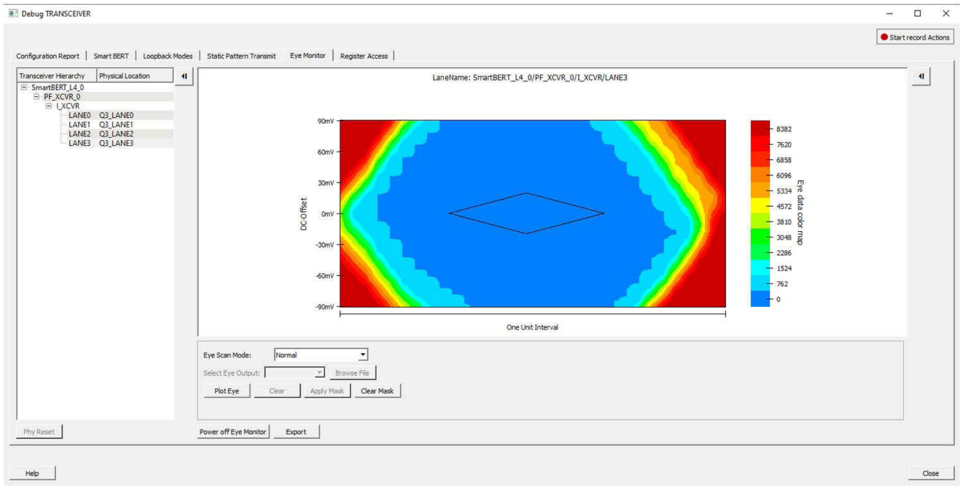
- Complete early power estimation by generating detailed power profiles based on logic, routing, clocking and I/O activity to enable accurate power budgeting and optimization
- Complete real-time power analysis, allowing you to fine tune your design for static and dynamic power efficiency

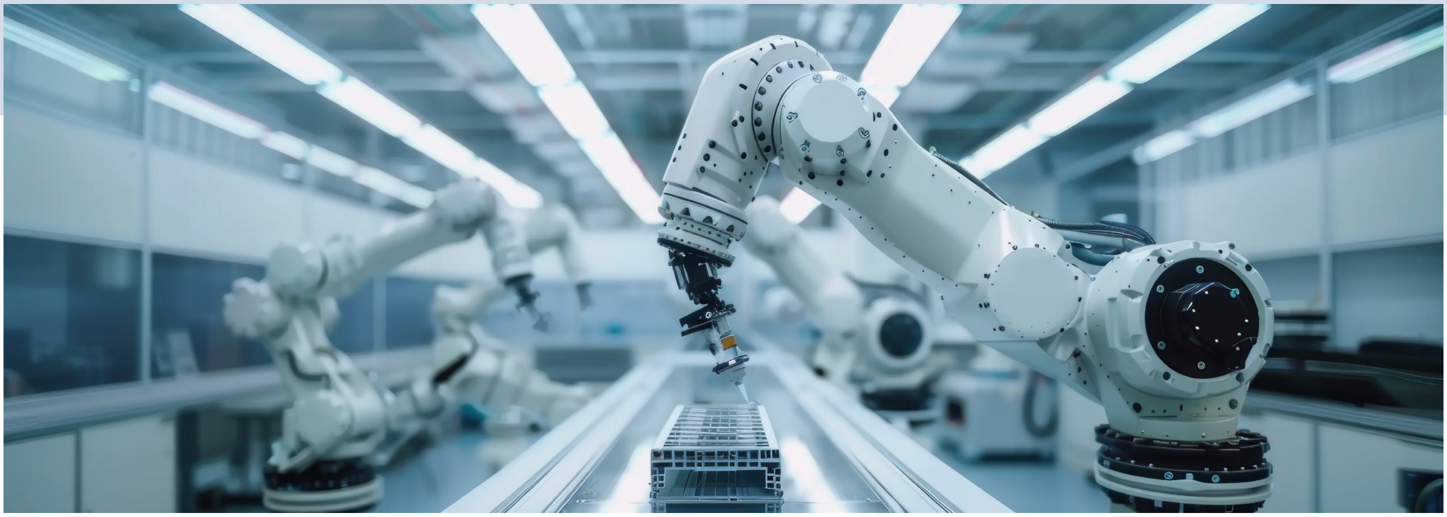
Advanced Debug

Design debugging is a critical phase of the FPGA design flow. SmartDebug simplifies this process by offering powerful verification and troubleshooting tools at the hardware level. It allows you to access and monitor various components of your design in real time without needing to make any modifications.

Identify ME is an Integrated Logic Analyzer (ILA)-based debugging tool which allows instrumentation directly from RTL or post-compile netlists and enables complex trigger mechanisms, automation and debugging at full speed with the integrated Identify RTL debugger.

Figure 3—SmartDebug Transceiver Eye Monitor GUI





IP Library

Our IP core tools supercharge your design process by offering a comprehensive suite of proven, optimized and user-friendly IP cores that are specifically crafted for our FPGAs and SoC FPGAs. Libero SoC Design Suite gives you seamless access to our in-house IP cores, which span critical functionalities like security, DSP, motor control and video and imaging.

To meet specialized market demands, CompanionCores IP cores offer targeted solutions to enhance our robust in-house offerings. We have partnered with industry leaders to provide IP cores tailored to AI/ML and aerospace and defense and other market segments.

Visit the IP Cores web page to learn more about our IP library.

Table 1: FPGA Support by Libero SoC Design Suite Version

Versions 2024.x to 12.0 (Inclusive)	Versions 11.9 and Earlier	Libero® IDE
PolarFire® FPGAs	SmartFusion 2 SoC FPGAs	ProASIC FPGAs
PolarFire SoC FPGAs	IGLOO 2 FPGAs	ProASIC Plus® FPGAs
RT PolarFire FPGAs	RTG4 FPGAs	RTAX FPGAs
RT PolarFire SoC FPGAs	SmartFusion FPGAs	RTSX FPGAs
SmartFusion® 2 SoC FPGAs	IGLOO FPGAs	Axcelerator FPGAs
IGLOO® 2 FPGAs		SX FPGAs
RTG4™ FPGAs	ProASIC® 3 FPGAs	eX FPGAs

Table 2: Silicon Device Support in Libero SoC Design Suite Versions 2024.x to 12.5

Product Family	Devices Supported	Silver (Free)	Gold	Platinum
PolarFire® SoC FPGAs	25K, 95K and 160K LEs (Non-S Devices)	✓	✓	✓
	Kit Parts: MPFS250T-FCG1152E, MPFS250T-FCVG484	✓	✓	✓
	250K LEs* (Non-S Devices)		✓	✓
	All Devices (Including S Devices)			✓
PolarFire FPGAs	100K LEs* (Non-S Devices)	✓	✓	✓
	200K LEs* (Non-S Devices)		✓	✓
	All Devices (Including S Devices)			✓
	Kit Parts: MPF300TS-1FCG1152I, MPF300TS-1FGG484I, MPF300TS-FGG484I, MPF300TS-1FCG1152E, MPF300TS-1FCG484E (Kit Parts)	✓	✓	✓
RTG4™ FPGAs	All Devices	✓	✓	✓
	5K to 25K LEs* (Except T2 Grade)	✓	✓	✓
SmartFusion® 2 and IGLOO® 2 FPGAs	5K LEs* (T2 Grade)			✓
	50K to 150K LEs* (Including Security)		✓	✓

*LEs = Logic Elements that feature a fully permutable four-input Lookup Table (LUT) with a dedicated carry chain and a separate flip-flop that can be used independently from the LUT

SMART | CONNECTED | SECURE

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